

Exploring the Impact of Lifestyle on Sleep and Productivity

Checkpoint 2



Data Analysis in Software Engineering 25/26

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Project structure

1

Independent Analysis of Mental Health and Productivity

Goal: study the effects of long-term lifestyle habits on productivity

Result: **20,000** rows and **8** features

Integrated Analysis: Coffee vs Power Nap with Lifestyle Profiles

Goal: study how intervention effects on alertness vary across matched lifestyle/productivity profiles.

Merged with K-Nearest Neighbors

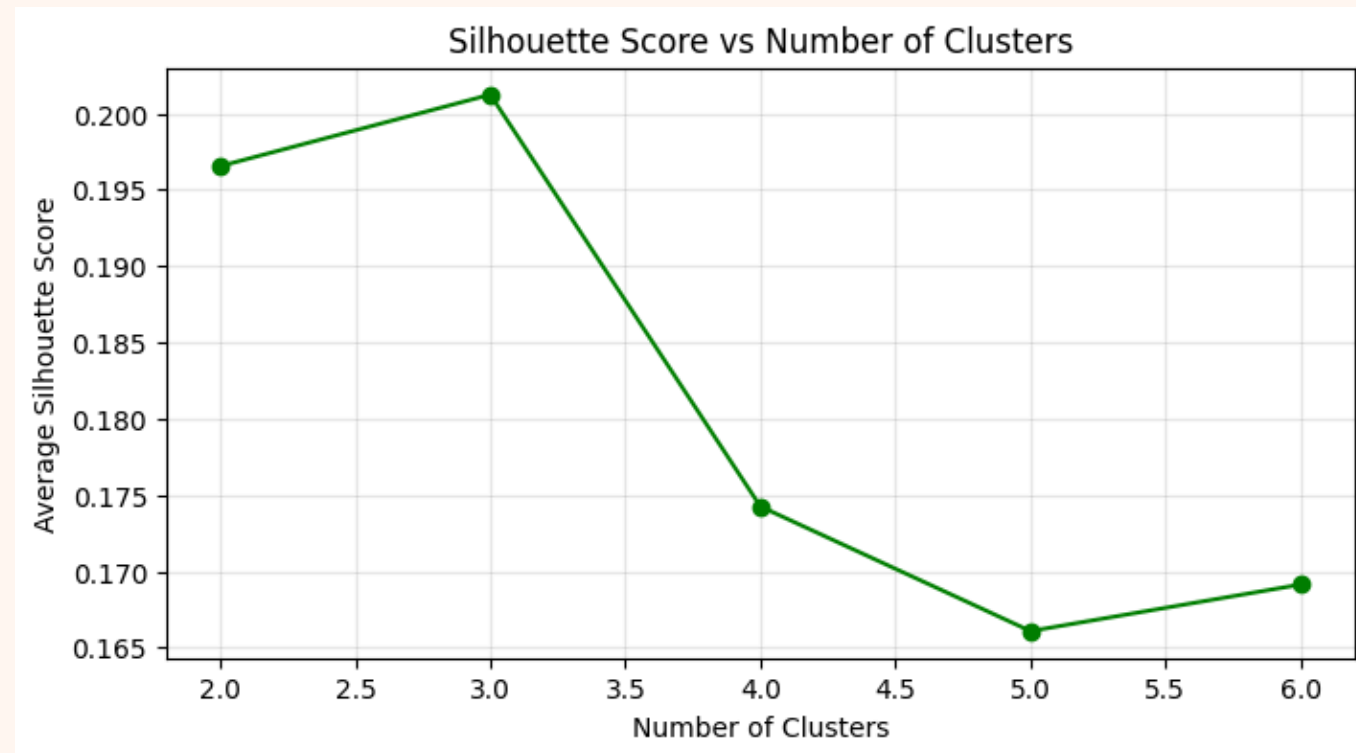
Result: **1,000** rows and **15** features

2

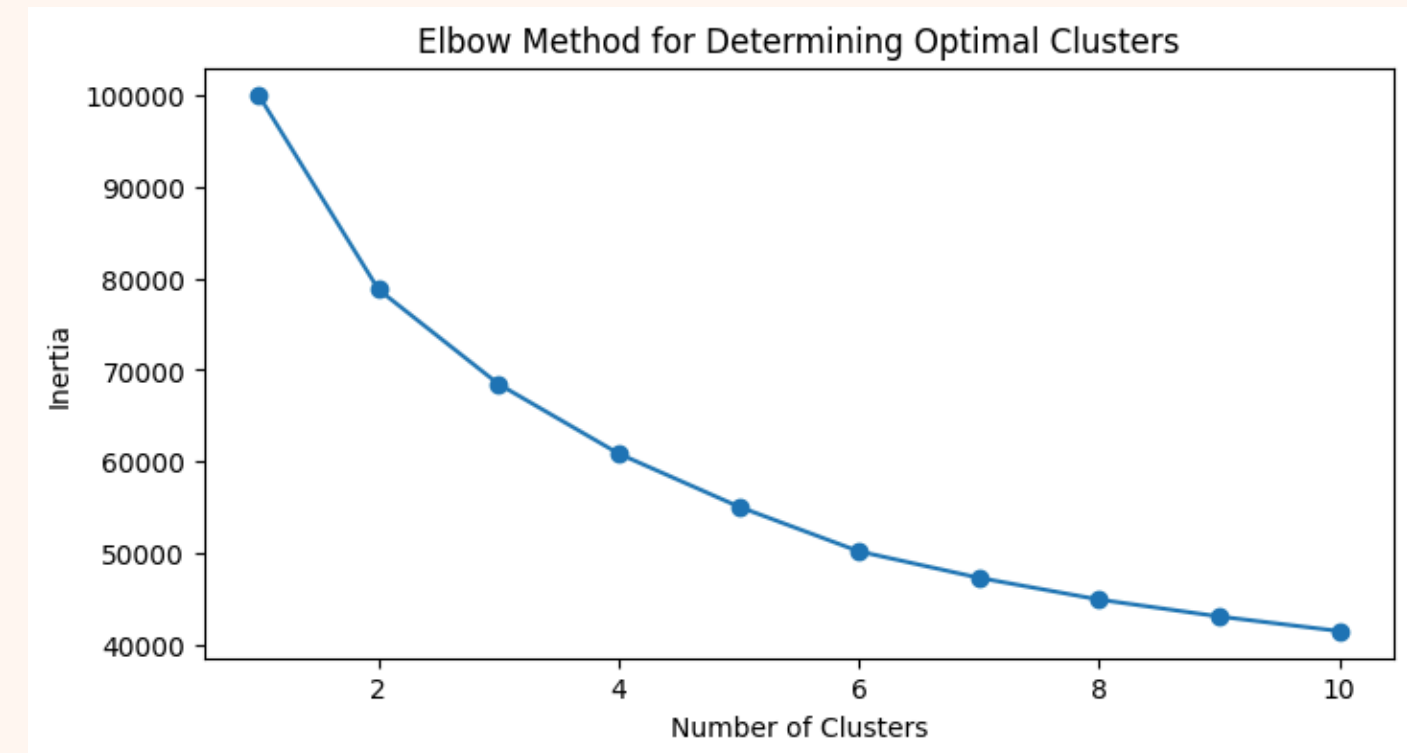
Main target: Predict productivity score and explain the most relevant factors

Part 1: Lifestyle Analysis

Unsupervised Learning



Silhouette Score

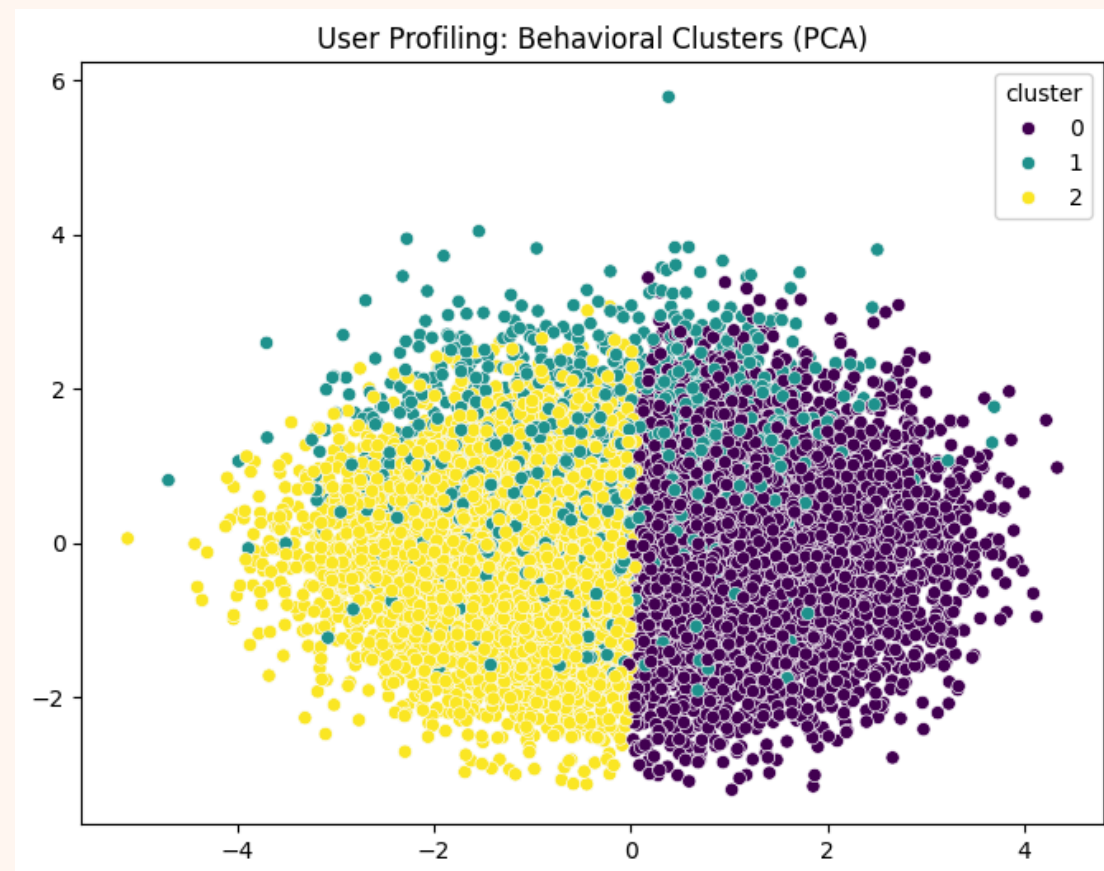


Elbow Method

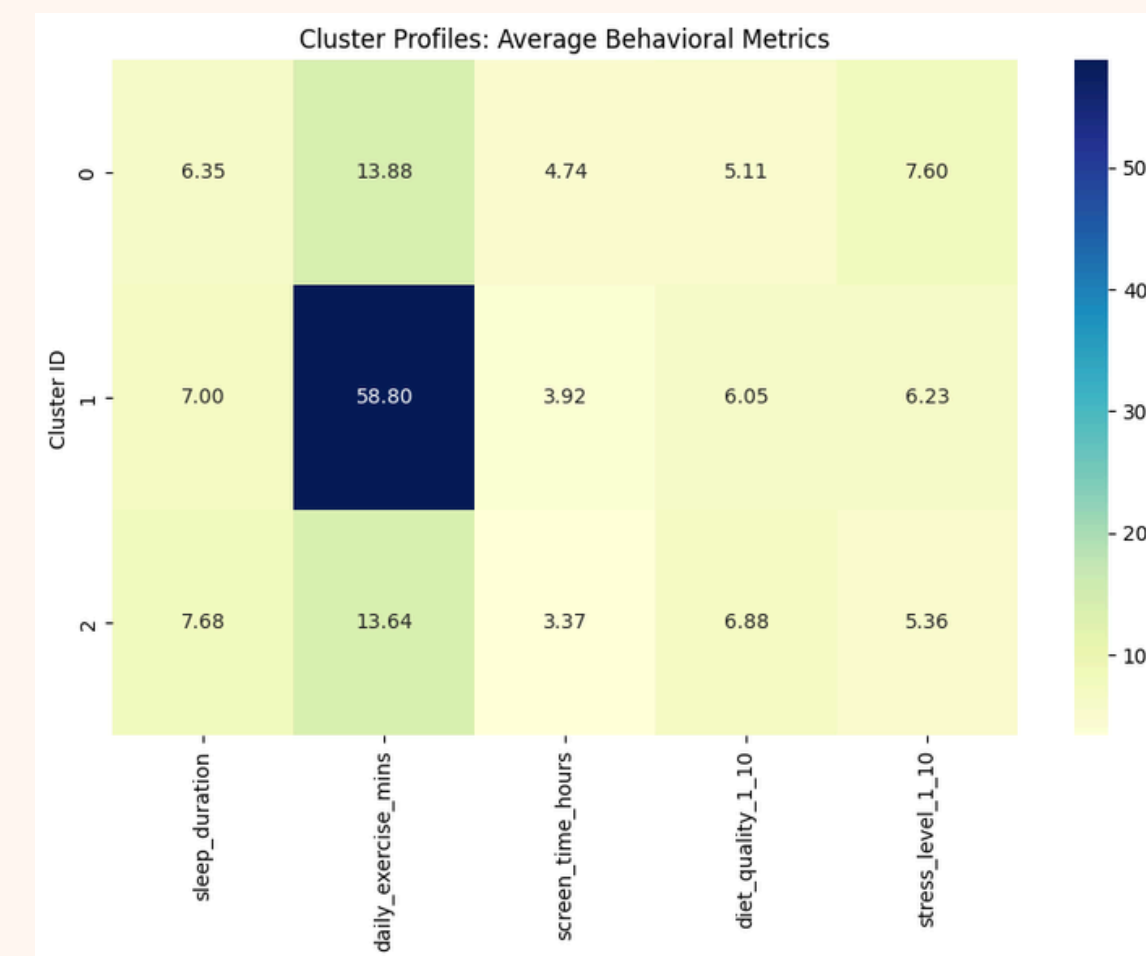
K=3 gave the best score

Part 1: Lifestyle Analysis

Unsupervised Learning



PCA

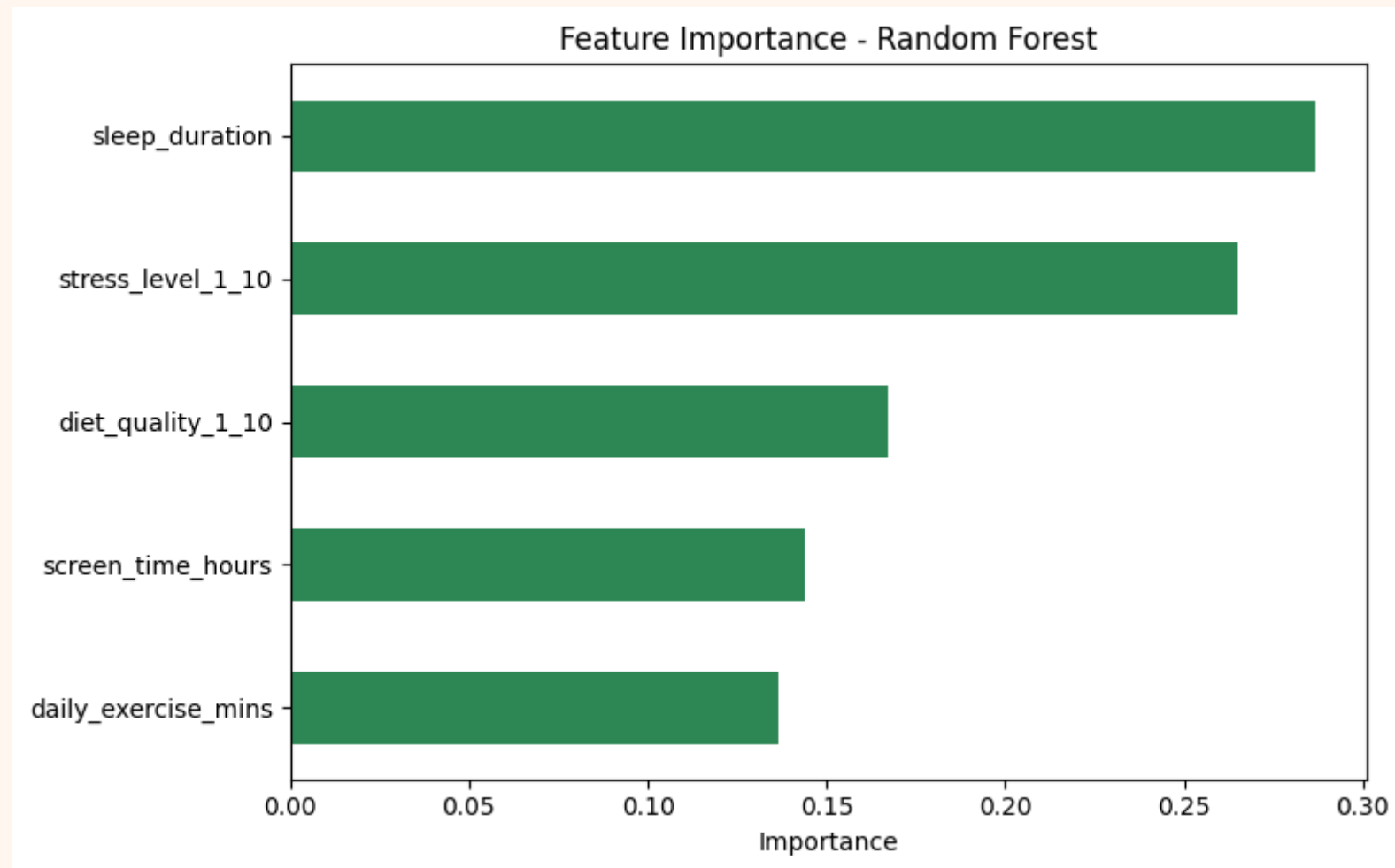


Cluster Profiling

PCA separates the users into three groups, and profiling shows one healthier, more active cluster

Part 1 Analysis

Supervised Learning



Random Forest

$R^2 = 0.5188$

MSE = 0.5666

We also did a Decision Tree and improved it with hyperparameter tuning using GridSearchCV with 5-fold cross-validation.

However, the best tree scores were worse:

$R^2 = 0.4934$

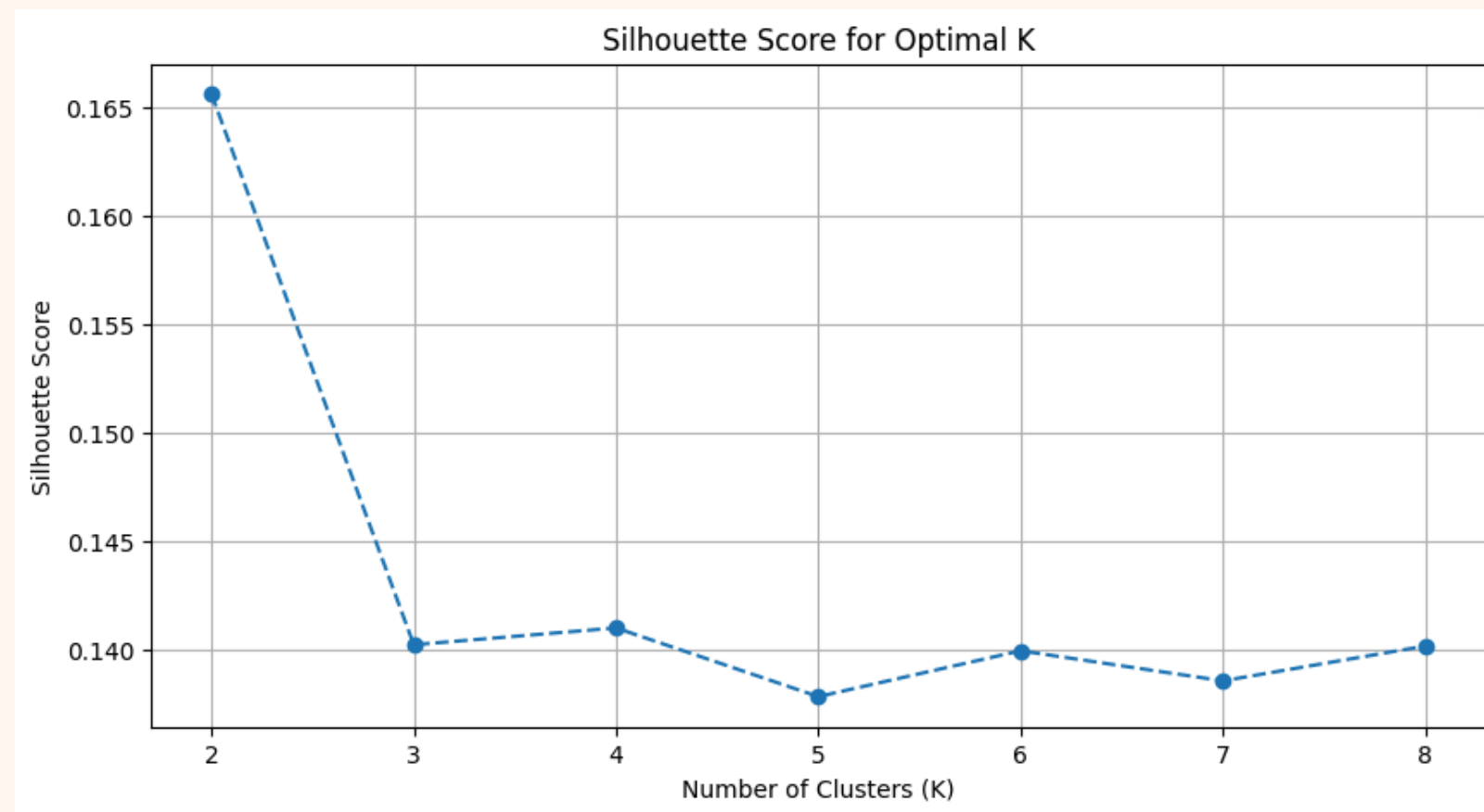
MSE = 0.6015

Preliminary results

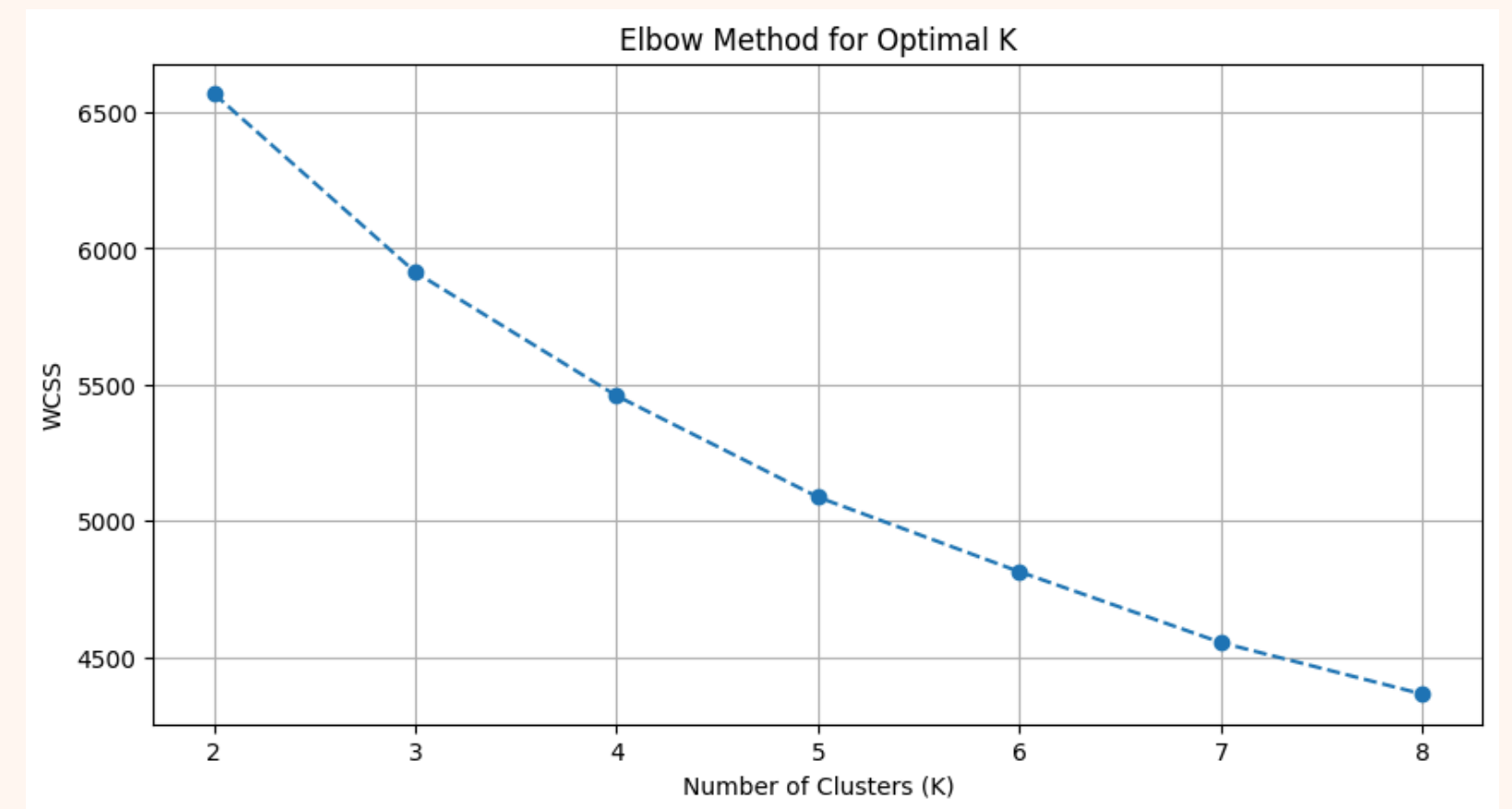
- The data suggests that lifestyle habits are **not random**; users naturally group into three distinct profiles, which means sleep, exercise, stress, and screen time tend to occur together in meaningful patterns.
- One cluster appears healthier and more active, so this likely represents a more balanced lifestyle profile associated with better sleep-related behavior.
- The Random Forest result suggests that lifestyle variables have moderate predictive power for productivity, but they do not explain everything, **so productivity is probably influenced by other factors too.**
- The fact that the tuned Decision Tree performed worse suggests that the relationship is not simple enough for a single interpretable tree, and that an ensemble model captures the patterns better.

Part 2: Merged Analysis

Unsupervised Learning



Silhouette Score

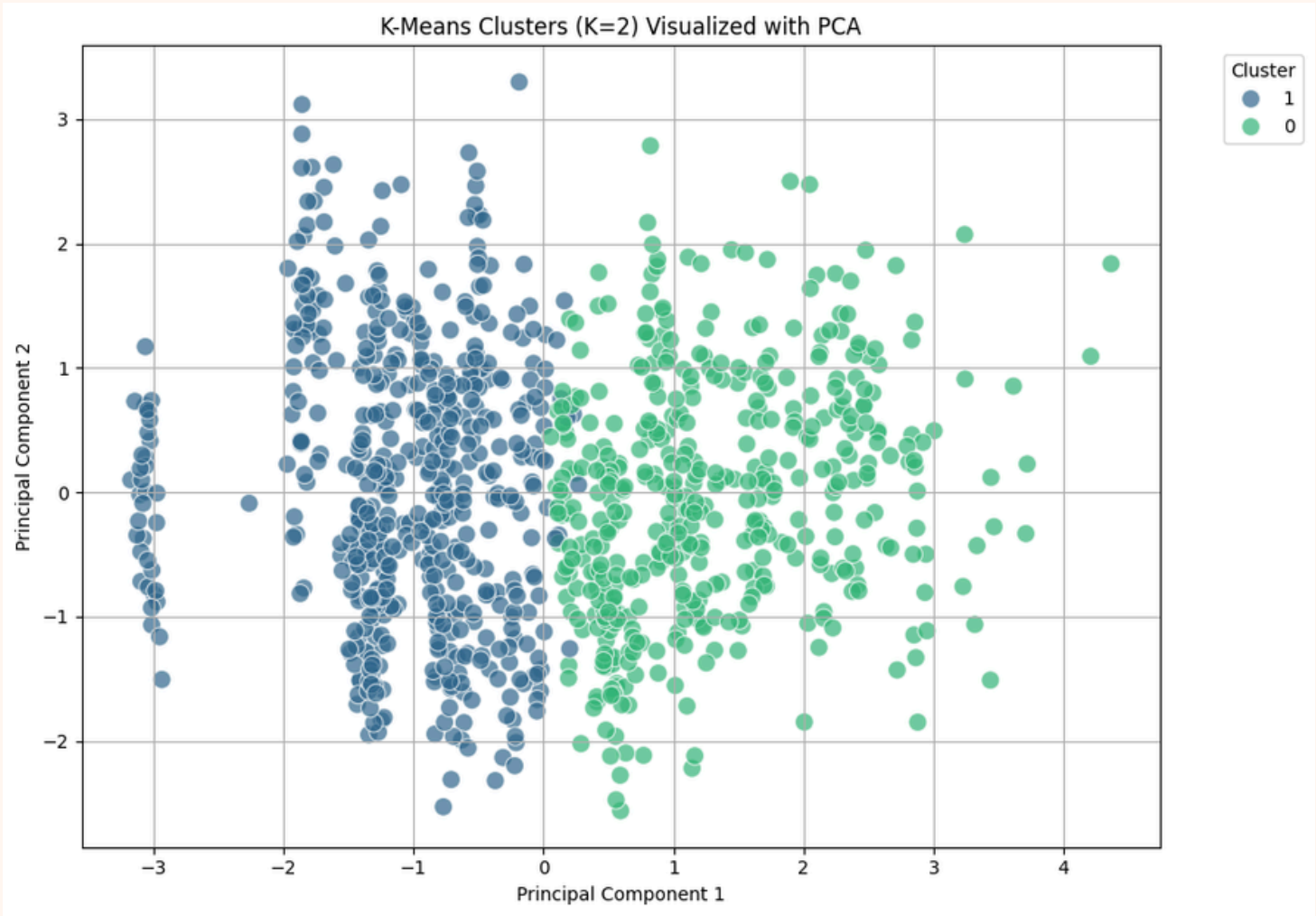


Elbow Method

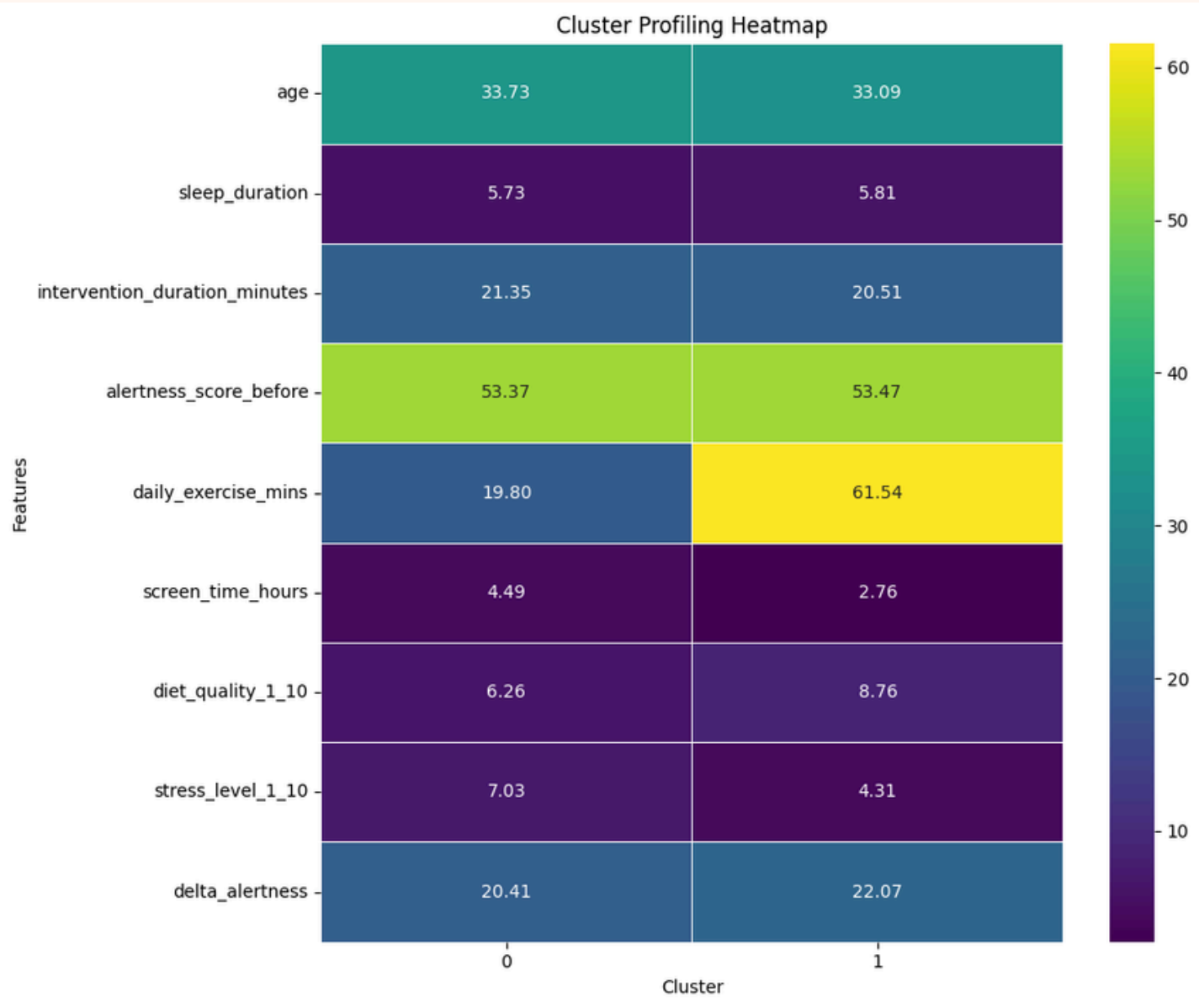
K=2 gave the best score

Part 2: Merged Analysis

Unsupervised Learning



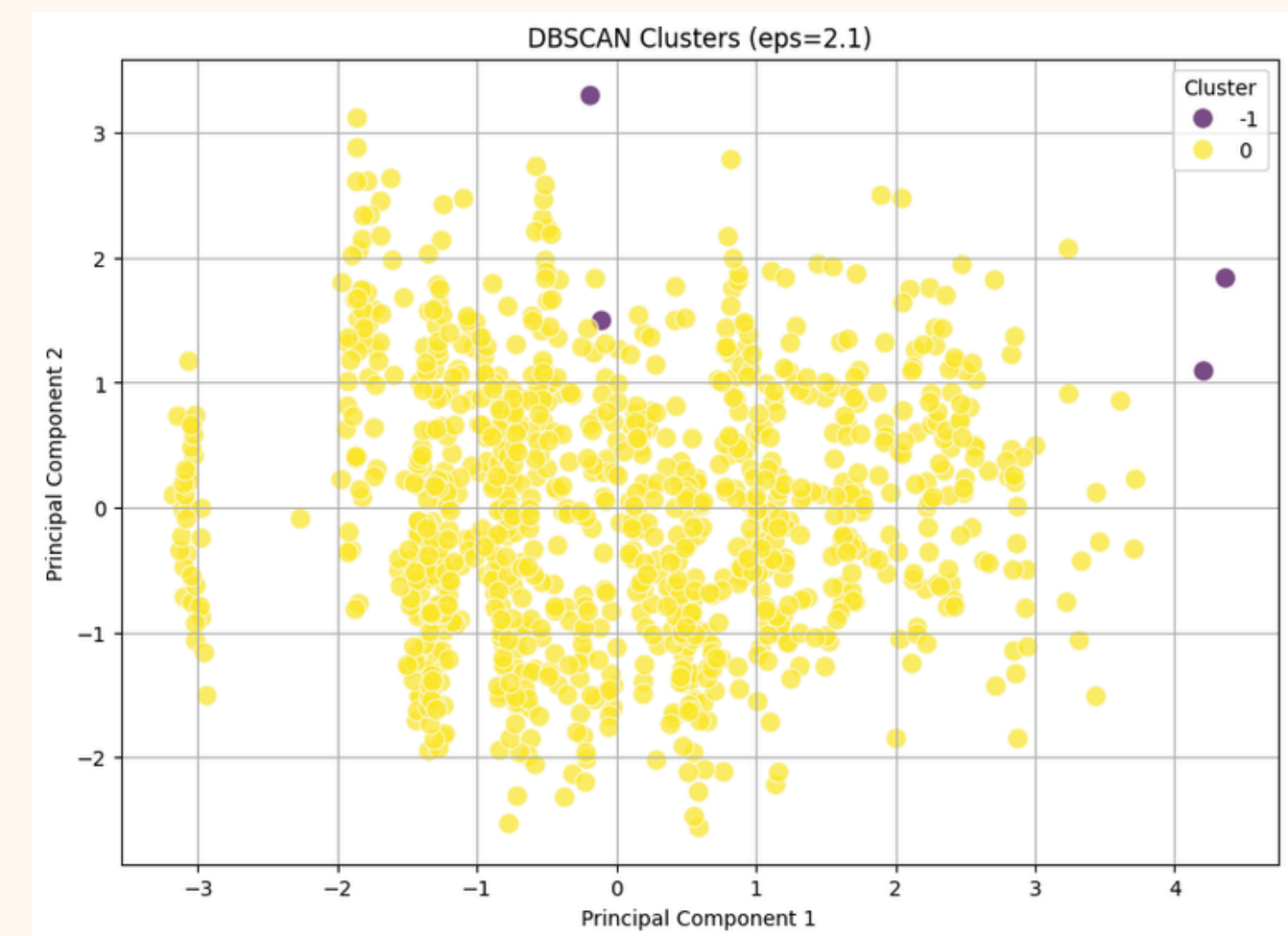
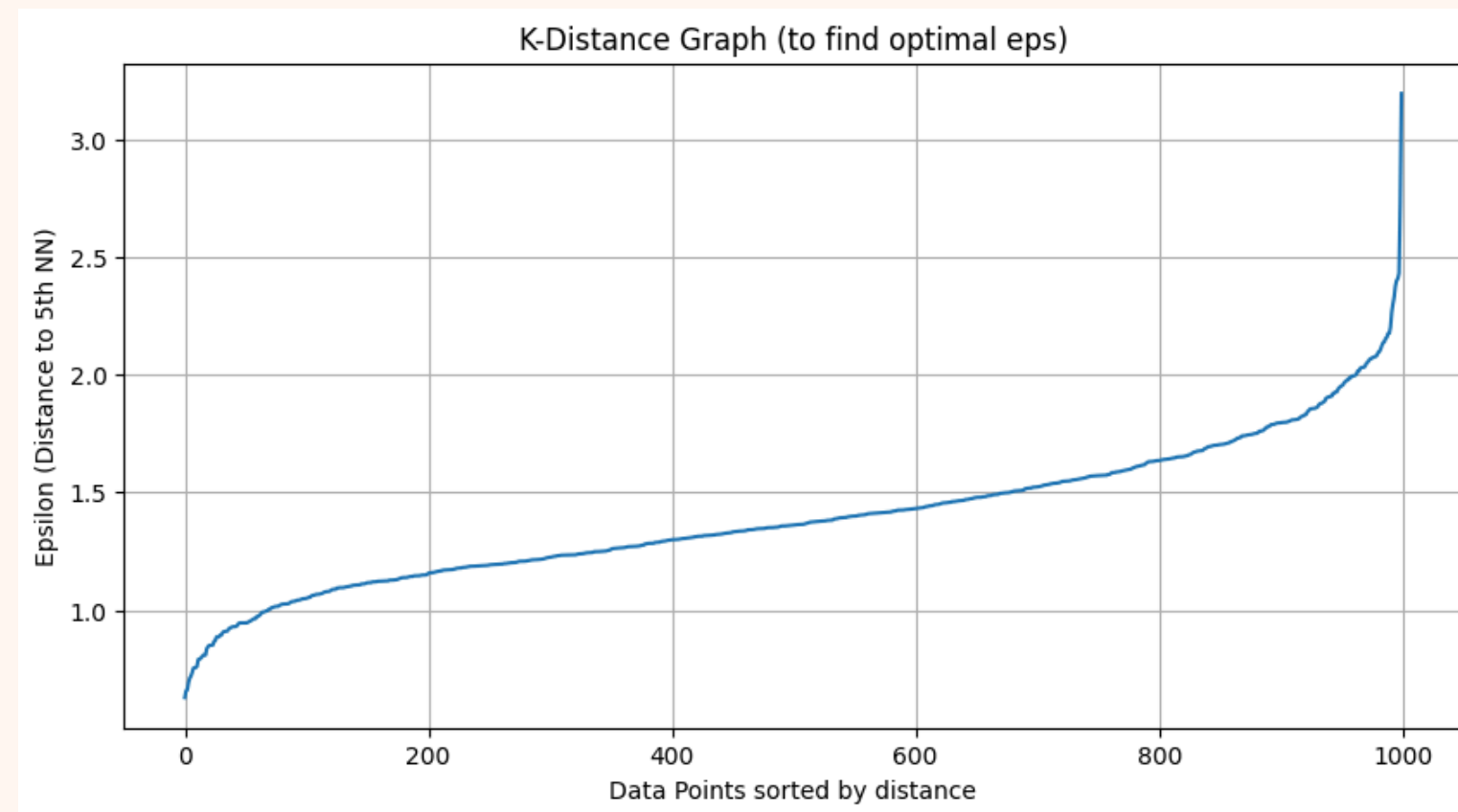
PCA Visualization



Profiling

Part 2: Merged Analysis

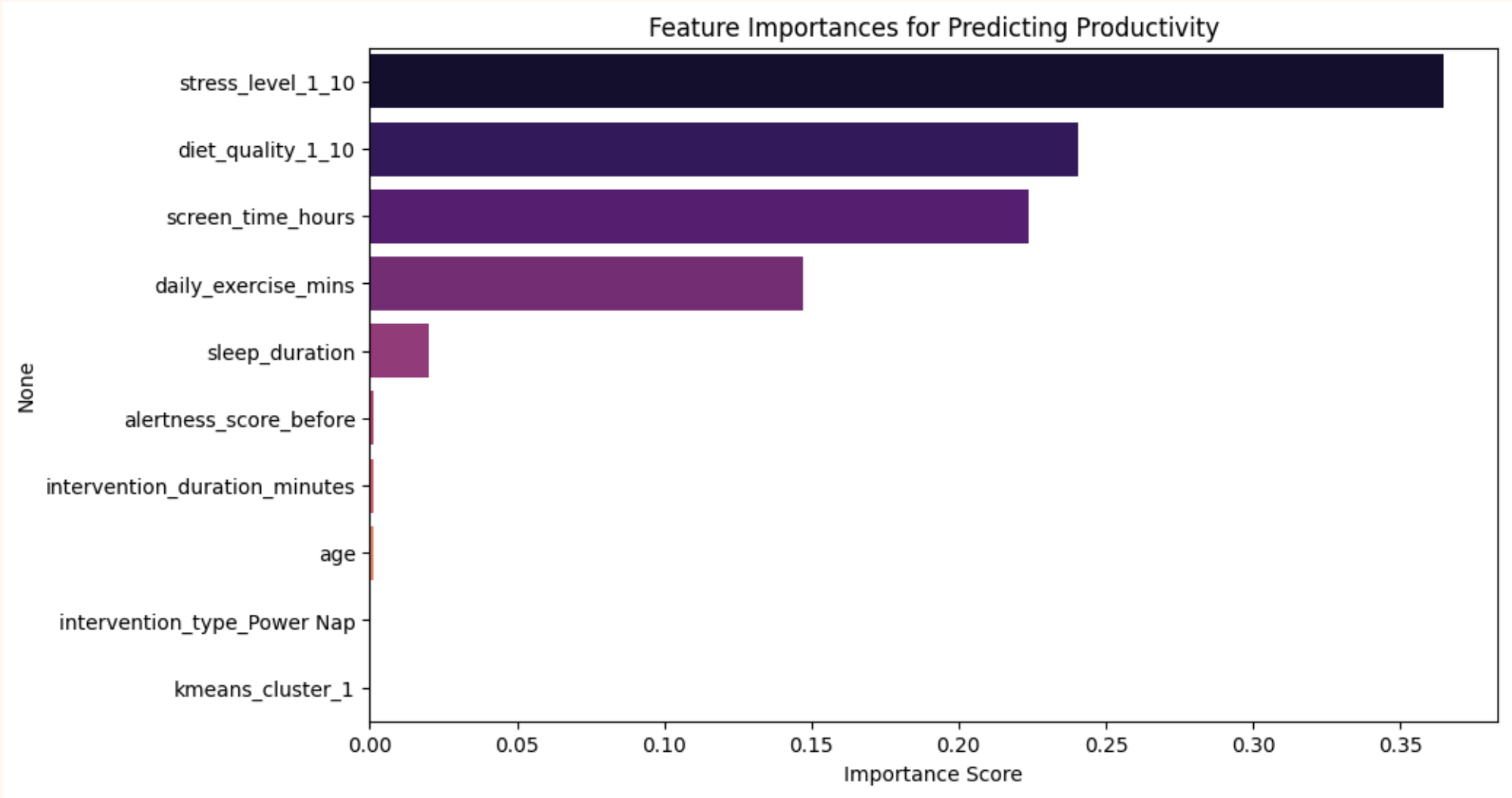
Unsupervised Learning



DBSCAN

Part 2: Merged Analysis

Supervised Learning



Decision tree

Decision Tree max_depth=7

R^2

Without Clustering: 0.8229

With Clustering: 0.8290

Improvement: 0.0060

Preliminary results

- Consistent, baseline habits, such as **good sleep and stress management**, matter more for overall **productivity** than “temporary boosters” coffee and power nap.
- The Decision Tree achieved an R^2 of 0.84 on the merged dataset, compared to 0.49 on the standalone dataset. This discrepancy could be attributed to target **leakage introduced by the KNN merge**, where productivity_rating was used as a matching feature, artificially inflating correlation between the merged lifestyle columns and the target. This a methodological limitation and motivates caution when interpreting the merged model's predictive power.

Next steps

Dataset 1 - Improving Accuracy

- Add interaction terms (e.g. sleep × exercise) to **capture non-linear relationships**
- **Feed cluster labels into the model** and measure R^2 impact

Dataset 2 - Fixing the Leakage and Analyze Clusters

- **Re-merge without productivity_rating** as a KNN matching feature and re-evaluate R^2
- **Split the merged dataset by cluster and compare productivity_rating gains** for Coffee and Power Nap within each group. This tests whether a "healthy" person benefits more or less from each intervention than an "unhealthy" one

Thank you!

